

Heart Attack – A Claims Point Of View

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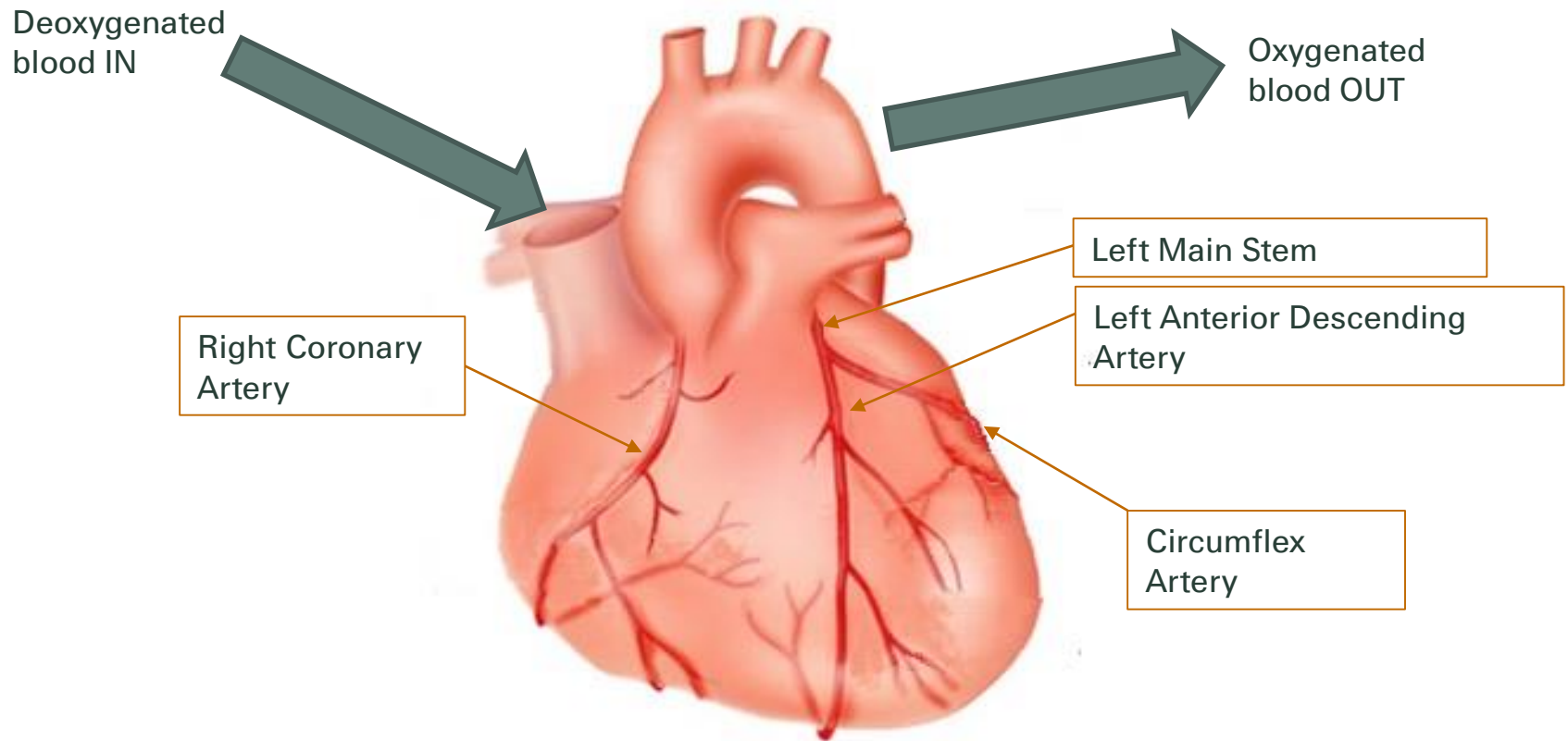


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Agenda

- Anatomy of the heart
- Facts & figures
- What is a heart attack?
- Key risk factors
- Diagnosis & treatment
- Definitions – history of change
- Evidence requirements & key considerations
- Recap quiz

Heart Anatomy



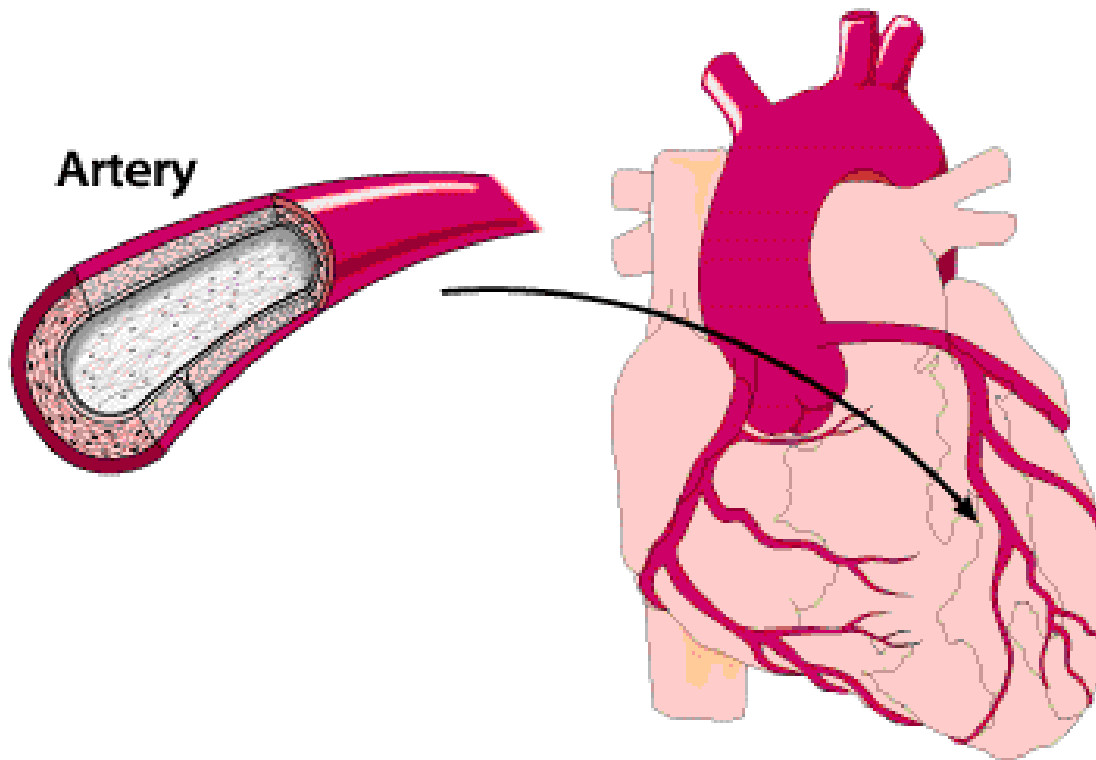
Heart Attack facts & figures



- 188,000 hospital visits each year due to heart attacks – that's one every three minutes *
- In the 1960's more than 7 out of 10 heart attacks in the UK were fatal. Today at least 7 out of 10 people survive *
- An estimated 915,000 people alive in the UK today have survived a heart attack (640,000 men and 275,000 women) *

* source – British Heart Foundation Sept 2016

What is a Heart Attack?

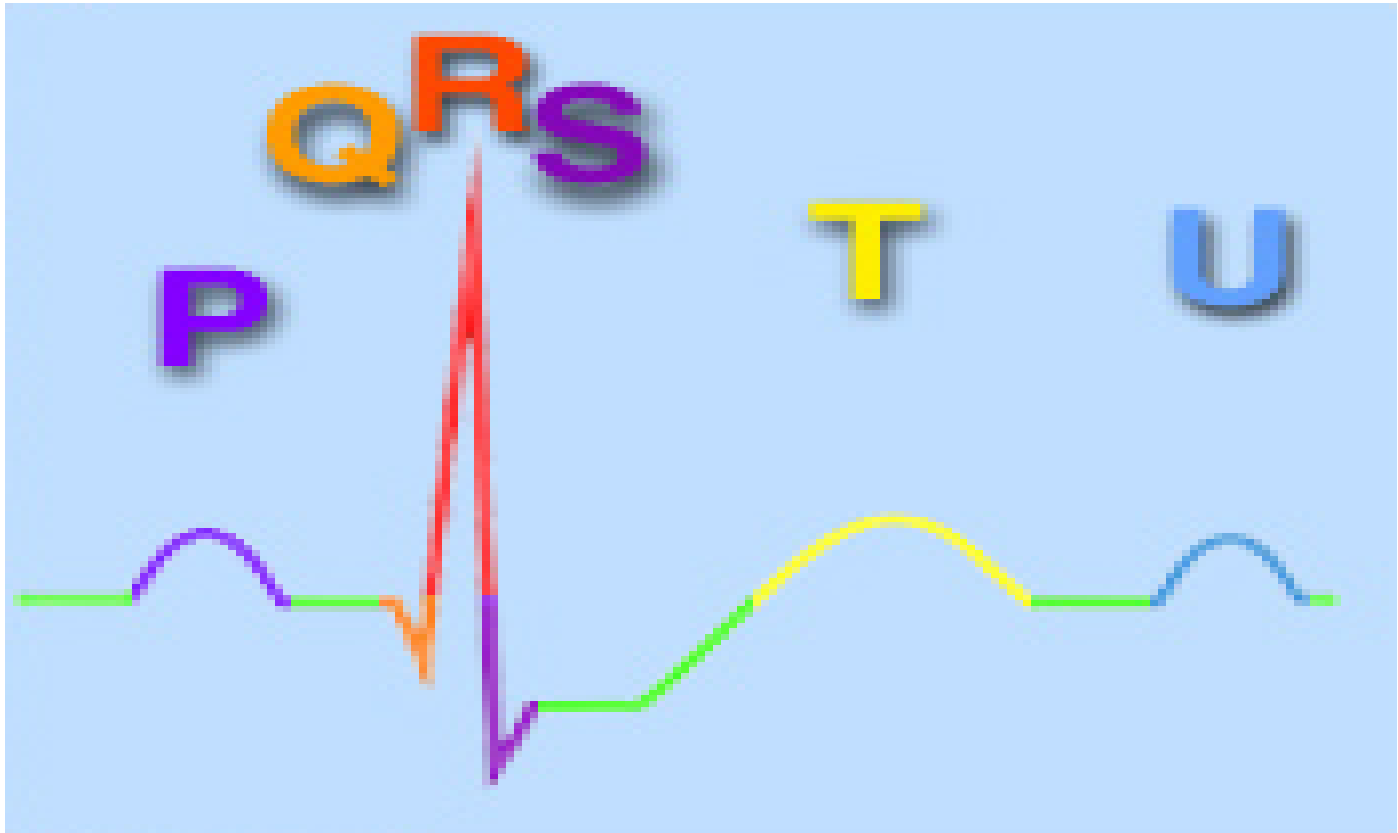


- Chest pain, typically radiating to left arm and jaw
- Ashen complexion
- Sweating
- Nausea
- Shortness of breath
- Loss of consciousness

Types of heart attack

- STEMI – Also referred to as a ST segment elevation myocardial infarction. This is the most serious type of heart attack where there is long interruption to the blood supply. This is caused by a total blockage to a coronary artery(s).
- NSTEMI – Also referred to as a Non-segment elevation myocardial infarction. Although still serious, it's not as serious as a STEMI as the blood supply to the heart is only partially blocked
- ACS – Acute Coronary Syndrome. This is the umbrella term for a heart attack but please be careful, unstable angina also comes under this title!
- Silent MI – This is a heart attack with few, if any symptoms. Most often an incidental finding when a patient is undergoing investigations for other coronary issues.

Normal ECG tracing



What are the key risk factors?

- Coronary Artery Disease
- Smoking
- Diet
- High blood pressure (hypertension)
- Diabetes mellitus
- Alcohol
- Age
- Gender
- Family history

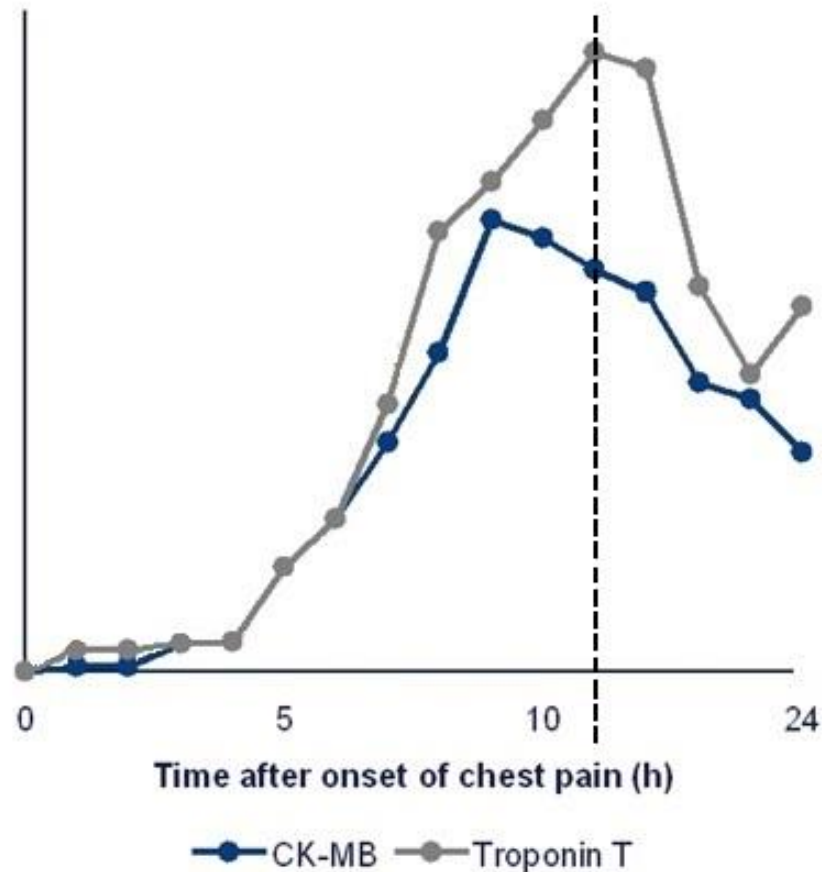
How is a Heart Attack diagnosed?

- Electrocardiogram (ECG) - a recording of the tiny electrical impulses used to make the heart contract. Electrodes are placed on the skin, and provide a report known as ECG 'tracings'. These are reviewed for abnormalities and can help to identify different locations of heart attacks and other cardiac syndromes.
- Blood test - a blood test will be arranged to review the level of enzymes linked to the presence of a heart attack, such as Troponin, or CK (Creatine Kinase) - and if these levels are elevated, a heart attack may be the cause.
- Angiogram - for this test a dye is used with scans to see the flow of blood within the heart and the coronary arteries to determine the location and extent of any occlusion or blockages.

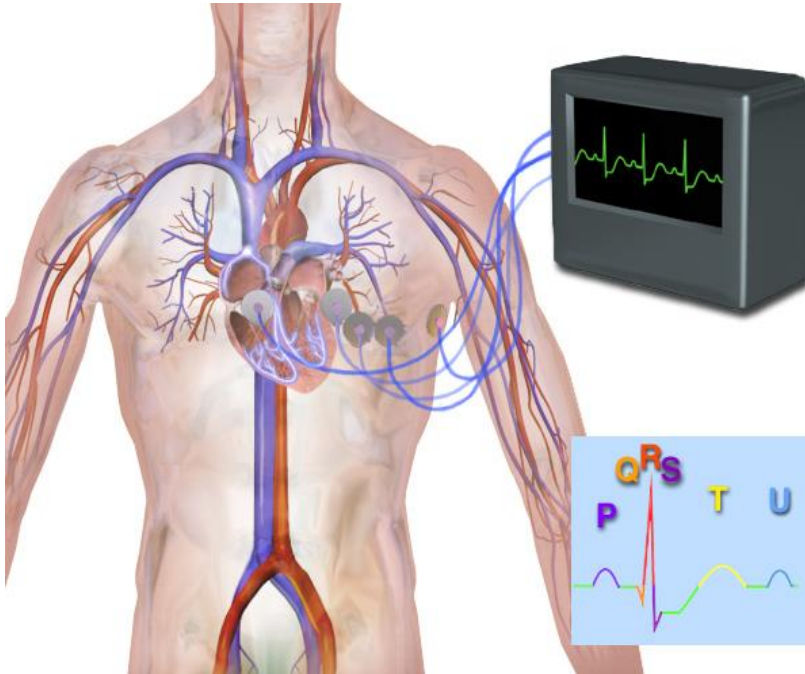
How is a Heart Attack diagnosed?

- Echocardiogram – This test is used to look at how blood flows through the heart chambers, heart valves and blood vessels. It's also used to detect damage done to the heart by a heart attack. There are several types but the most common is a transthoracic echocardiogram
- MRI – This is a type of scan which is very sensitive and is used to view the heart's structure and damage to your heart after a heart attack. It's also worth noting that they are also used to look at blood flow to the heart muscle that may cause chest pain and coronary heart disease.

How is a Heart Attack diagnosed?



How is a Heart Attack diagnosed?



Type of MI	Location	ECG Changes	Guidance
Anterior	Left ventricular wall	Q waves, ST segment & T wave changes V2-V6	Usually caused by LAD disease
Antero-lateral	Extends to the lateral wall	Q waves, ST segment & T wave changes I & Avl	
Antero-septal	Extends to the septum	Q waves extend V2-V4	
Inferior	Inferior part of the heart	Q waves, ST segment & T wave changes II, III & Avf	Usually caused by RCA or Circumflex disease
Infero-apical	Extension of an inferior MI to the apex	Changes throughout V5-V6	
Infero-lateral	Extension to the lateral part of the heart	Changes throughout I & Avl	
Posterior	Posterior part of the heart	Tall R segment in V1	Usually caused by RCA or Circumflex disease
Postero-lateral	Extends to the lateral portion of the heart	Changes throughout I & Avl	
Subendocardial	Does not penetrate through the full thickness of the myocardium	Non-specific T wave changes, no Q waves	More favourable mortality

How is a Heart Attack treated?

- **Thrombolysis** - early intervention during the development of a cardiac event using clot busting drugs which prevent ischaemia (i.e. may avert a heart attack, which is an important consideration in the assessment of a claim).
- **Medication** - statins can help reduce cholesterol thereby reducing a risk factor for further heart attacks; anti-clotting drugs can also help fight against further heart blockage.
- **Lifestyle advice** - changes in diet, smoking/alcohol habits and an increase in exercise to reduce future risk of further cardiac events.
- **Angioplasty (also called percutaneous coronary intervention (PCI))** - the process of improving blood flow through narrowed coronary arteries. This is done by placing expanding metallic stents into coronary arteries and then inflating them so as to squash the cholesterol plaques and improve blood flow.
- **Coronary artery bypass grafting (CABG)** - the process of bypassing coronary arteries that are narrowed as a result of coronary artery disease in order to improve blood flow.

Definitions bit

2014 – ABI – Heart Attack (of specified severity):

Death of heart muscle, due to inadequate blood supply, that has resulted in all of the following evidence of acute myocardial infarction:

- Typical clinical symptoms (for example, characteristic chest pain).
- New characteristic electrocardiographic changes.
- The characteristic rise of cardiac enzymes or Troponins recorded at the following levels or higher;
 - Troponin T > 200 ng/L (0.2 ng/ml or 0.2 ug/L)
 - Troponin I > 500 ng/L (0.5 ng/ml or 0.5 ug/L)

The evidence must show a definite acute myocardial infarction.

For the above definition, the following are not covered:

- Other acute coronary syndromes
- **Angina without myocardial infarction**

Some thoughts about question wording

- Do we ask for the right information?
- Do we ask for information that isn't particularly relevant?
- Do we ask the question in the right way?

- 'Please give details of all tests and investigations and their results'
 - non specific
 - 'all' tests and investigations?
 - open to what the doctor wants to tell us

Some thoughts about question wording

➤ Make the questions more specific and directed

- 'please confirm the symptoms and time/date of these symptoms'
- 'details of ECG changes throughout the stay in hospital'
- 'please confirm the date and time of all cardiac enzyme tests, the results and the lab reference ranges'
- 'what were the results of angiography and was any treatment given'
- 'was echocardiography performed as if so, was there wall motion abnormality. Please give details'

What else could it be?

The criteria are fulfilled but it's a different diagnosis

➤ Viral myocarditis

- chest pain associated with fever, possibly no chest pain
- can give a rise in troponin and ECG changes

➤ Tachycardia

- racing heart and possibly chest pain
- can give a rise in troponin and ECG changes

➤ Cardiac arrest

- massively significant event – critical by its very nature
- can give a rise in troponin and ECG changes
- is there death of heart muscle?

Recap Quiz

- **What causes a heart attack?**

- ☐ A. Too much cholesterol that can clog your arteries
- ☐ B. A blood clot that blocks blood flow to the heart
- ☐ C. A spasm in an artery that supplies blood to the heart (coronary artery)
- ☐ D. All of the above

- **What are the risk factors you can't control?**

- ☐ A. Age
- ☐ B. Gender
- ☐ C. Family history
- ☐ D. Race
- ☐ E. All of the above

Recap Quiz

- **Stress can add to your risk of heart disease.**

- ☐ A. True
- ☐ B. False

- **You can't miss signs of a heart attack.**

- ☐ A. True
- ☐ B. False

- **You can prevent a heart attack.**

- ☐ A. True
- ☐ B. False

Thank you



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